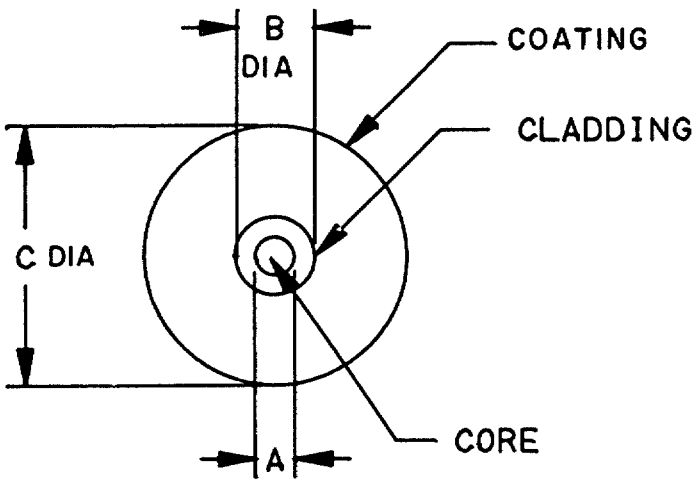
MILITARY SPECIFICATION SHEET

FIBER, OPTICAL, 100/140 µm, RADIATION HARDENED (METRIC)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the optical fibers described herein shall consist of this specification sheet and the latest issue of DOD-F-49291.

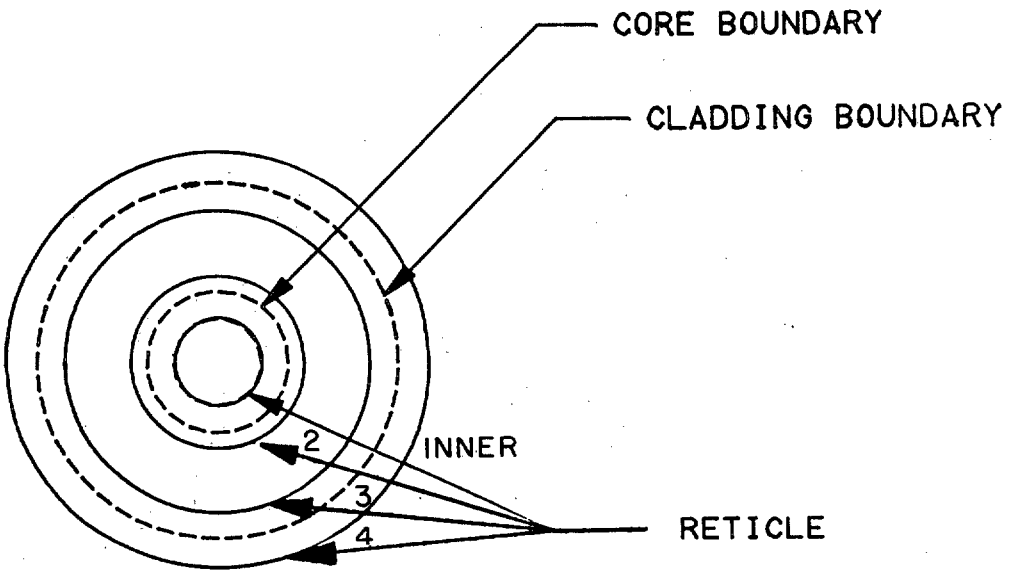


Fiber index	Part number	Dimension (µm)		
		A	B	C
Graded	D49291/2-01	100 ± 3	140 ± 2	250 ± 15
Quasi graded	D49291/2-02	100 ± 3	140 ± 2	250 ± 25

NOTE: Dimensions are in micrometers.

FIGURE 1. Dimensions and configuration of primary coated optical fiber construction.

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Circle	Diameter of circle
Inner	97 µm
Second	103 µm
Third	138 µm
Fourth	142 µm

NOTE: Dimensions are in micrometers.

FIGURE 2. Tolerance fields.

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REQUIREMENTS:

Classification:
Type: I
Class: 1 or 2 (see part number)
Composition: A
Style: 1

Dimensions and configuration:
Diameter: See figures 1 and 2.
Ovality:
Core: 5 percent
Cladding: 4 percent
Coating: 10 percent
Offset:
Core-to-cladding: 6 percent
Fiber-to-coating: 14 percent
Maximum percent of coating diameter change at the splice points: Not applicable.
Continuous lengths: 1100 meters minimum.

Splices: Not allowed.

Tensile strength (proof test): 690 MPa.

Fiber mass/unit length (kg/km): 1 kg/km maximum.

Attenuation rate: 3.5 dB/km, 1290 nm ±10 nm, 10 nm full width half maximum (FWHM); 2.5 dB/km, 850 nm ±10 nm, 10 nm FWHM.
After nuclear event: Maximum allowable degradation is 2 dB at -46°C, 1 dB at +25°C, 1 dB at +71°C.

Numerical aperture: 0.29 ±0.02 at 850 nanometers.

Bandwidth:
Graded index:
200 MHz·km (850 nm ±10 nm)
200 MHz·km (1290 nm ±10 nm)
Quasi-graded index:
100 MHz·km (850 nm ±10 nm)
100 MHz·km (1290 nm ±10 nm)

Fluid immersion: Not Applicable.

Maximum percentage change in coating diameter due to immersion: 10 percent.

Temperature range:
Operating: -46°C to +71°C.
Storage: -57°C to +85°C.

Fungus test: Applicable.

Nuclear radiation resistance: The nuclear radiation resistance characteristics of this optical fiber are classified and are contained in an appendix to this specification. Application to receive this appendix must be made through the US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703-5016. Information concerning security clearance classification and "need to know" must be detailed in the request.

Part number (see figure 1): D49291/2-01
D49291/2-02

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Qualification inspection shall consist of performing the inspections and optical tests specified in Table I herein, in the sequence shown.

Group A inspection shall consist of performing the inspections and optical tests specified in Table II herein, in the sequence shown.

Group B inspection shall consist of performing the inspections and optical tests specified in Table III herein, in the sequence shown.

Group C inspection shall consist of performing the inspections and optical tests specified in Table IV herein, in the sequence shown.

TABLE I. Qualification inspection.

Inspection	Specimen length	Sample quantity
Group I		
Visual and mechanical	1 km	10 units
Attenuation rate	1 km	10 units
Numerical Aperture	2 m	10 units
Core diameter	2 m	10 units
Bandwidth by time domain	1 km	10 units
Group II		
Proof test	1 km	5 units
Mechanical strippability	2 m	5 units
Group III		
Temperature cycling	1 km	5 units
Temperature shock	1 km	5 units
Temperature-humidity cycling	1 km	3 units
Dimensional stability	5 m	3 units
Nuclear radiation resistance	50 m min.	9 units 1/
Fungus	5 m	2 units
Life aging	1 km	2 units

1/ 3 units each from 3 fibers, 3 units for each test temperature, 9 units total.

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TABLE II. Group A inspection.

Inspection	Specimen length
Visual and mechanical	1 km
Attenuation rate	1 km
Numerical aperture	2 m
Core diameter	2 m
Bandwidth by time domain	1 km

TABLE III. Group B inspection.

Inspection	Specimen length
Proof test	1 km
Mechanical strippability	2 m

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TABLE IV. Group C inspection.

Inspection	Specimen length	Sample quantity
Subgroup I (3 month interval)		
Temperature cycling	1 km	4 units
Subgroup II (6 month interval)		
Temperature shock	1 km	2 units
Temperature-Humidity cycling	1 km	2 units
Subgroup III (12 month interval)		
Dimensional stability	5 m	2 units
Nuclear radiation resistance	50 m min.	6 units 1/
Fungus	5 m	2 units
Life aging	1 km	2 units

1/ 3 units each from 2 fibers, 2 units for each test temperatures 6 units total.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI, SC
Navy - AS
Air Force - 11
DLA - ES

User activity:
Air Force - 17

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6010-0013-2)

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